

1-task

- `INSERT INTO flights (flight number, departure airport, arrival airport, departure_time, arrival_time)`
- `SELECT`
- `'FL-' || floor(random() * 8999 + 1000)::text,`
- `'ALA',`
- `'TSE',`
- `now() - (random() * 365 || ' days')::interval,`
- `now() - (random() * 365 || ' days')::interval + interval '2 hours'`
- `FROM generate_series(1, 200)`

The screenshot shows a PostgreSQL query editor with the following SQL code:

```
1 INSERT INTO flights (flight_number, departure_airport, arrival_airport, departure_time, arrival_time)
2 SELECT
3     'FL-' || floor(random() * 8999 + 1000)::text,
4     'ALA',
5     'TSE',
6     now() - (random() * 365 || ' days')::interval,
7     now() - (random() * 365 || ' days')::interval + interval '2 hours'
8 FROM generate_series(1, 200);
```

The query has been executed successfully, as indicated by the status bar at the bottom: "Query returned successfully in 189 msec."

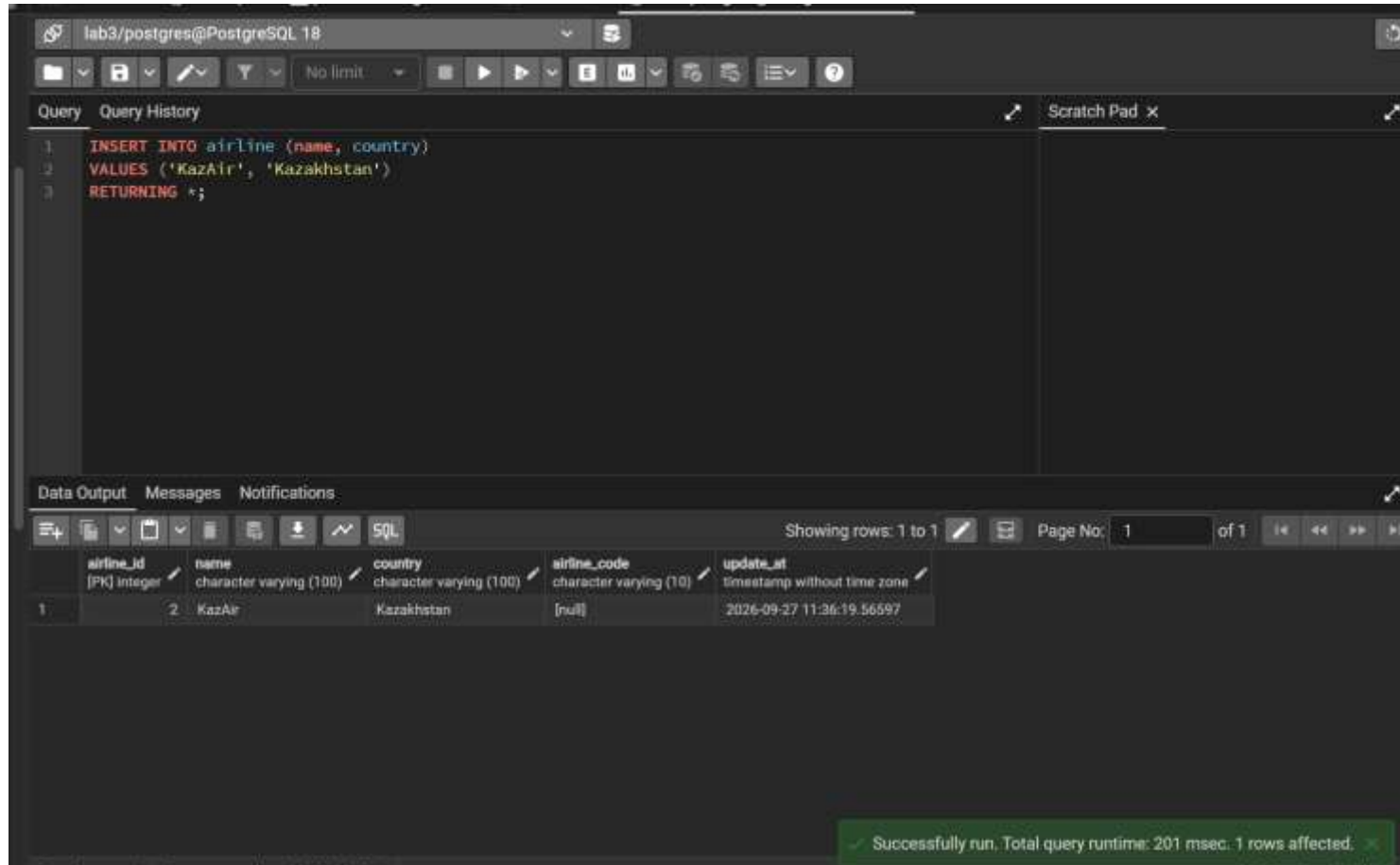
The screenshot shows a PostgreSQL query editor with the following SQL code:

```
1
2 SELECT * FROM flights LIMIT 10;
```

The query has been executed successfully, as indicated by the status bar at the bottom: "Successfully run. Total query runtime: 172 msec. 10 rows affected."

flight_id	flight_number	departure_airport	arrival_airport	departure_time	arrival_time
1	FL-3670	ALA	TSE	2025-11-25 22:24:17.1221	2026-04-14 23:35:42.209723
2	FL-7272	ALA	TSE	2026-01-23 12:52:47.435445	2026-04-26 21:07:21.827977
3	FL-9452	ALA	TSE	2025-10-02 04:06:17.312281	2025-10-02 14:31:56.891153
4	FL-7179	ALA	TSE	2026-06-13 02:08:57.133437	2026-07-16 07:10:10.14201
5	FL-6550	ALA	TSE	2026-02-10 10:36:34.398547	2026-05-04 23:41:28.421992
6	FL-9204	ALA	TSE	2026-02-15 19:33:39.787997	2026-06-13 13:24:13.352628
7	FL-3899	ALA	TSE	2025-10-13 11:15:54.817969	2026-03-20 22:58:40.254372
8	FL-4657	ALA	TSE	2026-01-27 08:53:49.091252	2026-05-02 20:31:55.121676
9	FL-9044	ALA	TSE	2025-12-29 23:47:1	

2-task



The screenshot displays a PostgreSQL client window with the title 'lab3/postgres@PostgreSQL 18'. The interface includes a toolbar with icons for file operations, query execution, and settings. The 'Query' tab is active, showing the following SQL statement:

```
1 INSERT INTO airline (name, country)
2 VALUES ('KazAir', 'Kazakhstan')
3 RETURNING *;
```

The 'Data Output' tab is also active, displaying the results of the query in a table format. The table has five columns: 'airline_id' (integer, primary key), 'name', 'country', 'airline_code', and 'update_at' (timestamp without time zone). The results show one row with the following values:

airline_id	name	country	airline_code	update_at
2	KazAir	Kazakhstan	[null]	2026-09-27 11:36:19.56597

At the bottom of the window, a green status bar indicates: 'Successfully run. Total query runtime: 201 msec. 1 rows affected.'

3-task

The screenshot shows a PostgreSQL query editor interface. The top toolbar includes icons for saving, running, and other database operations. The main query area contains the following SQL statement:

```
UPDATE airline
SET country = 'Turkey'
WHERE name = 'KazAir'
RETURNING *;
```

Below the query editor, the 'Data Output' tab is active, displaying the results of the query in a table. The table has five columns: `airline_id` (integer), `name` (character varying), `country` (character varying), `airline_code` (character varying), and `update_at` (timestamp without time zone). The results show two rows, both with `name` 'KazAir' and `country` 'Turkey'.

airline_id	name	country	airline_code	update_at
1	KazAir	Turkey	[null]	2025-09-27 11:26:06.523291
2	KazAir	Turkey	[null]	2025-09-27 11:36:19.56597

At the bottom of the interface, a green status bar indicates: 'Successfully run. Total query runtime: 279 msec. 2 rows affected.' The bottom status bar also shows 'Total rows: 2', 'Query complete 00:00:00.279', and 'CRLF Ln 4, Col 13'.

4-task

The screenshot displays a PostgreSQL client window with the following components:

- Query Editor:** Contains an SQL `INSERT` statement:

```
1 INSERT INTO airline (name, country)
2 VALUES
3     ('AirEasy', 'France'),
4     ('FlyHigh', 'Brazil'),
5     ('FlyFly', 'Poland')
6 RETURNING *;
```
- Data Output Panel:** Shows the result of the query as a table with 5 columns: `airline_id` (PK integer), `name` (character varying (100)), `country` (character varying (100)), `airline_code` (character varying (10)), and `update_at` (timestamp without time zone). The table contains 3 rows of data.
- Status Bar:** Displays "Total rows: 3" and "Query complete: 00:00:00.229".
- Message Panel:** Shows a green notification: "Successfully run. Total query runtime: 229 msec. 3 rows affected."

airline_id	name	country	airline_code	update_at
1	AirEasy	France	[null]	2026-09-27 11:37:06.645018
2	FlyHigh	Brazil	[null]	2026-09-27 11:37:06.645018
3	FlyFly	Poland	[null]	2026-09-27 11:37:06.645018

5-task

The screenshot shows a PostgreSQL IDE interface. The top toolbar includes icons for file operations, query execution, and settings. The main query editor contains the following SQL code:

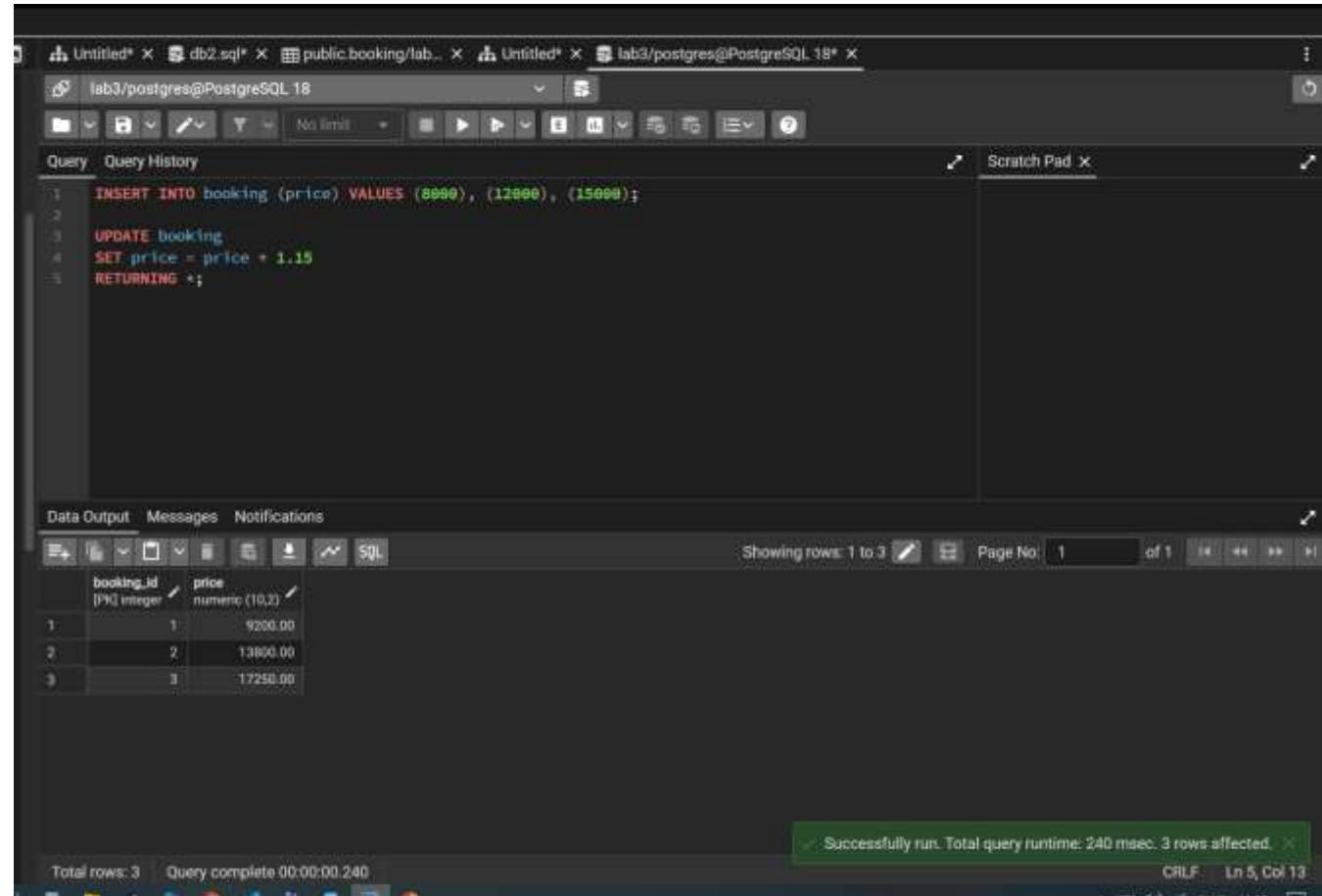
```
1 DELETE FROM flights
2 WHERE EXTRACT(YEAR FROM arrival_time) = 2024
3 RETURNING *;
```

Below the query editor, the 'Data Output' tab is active, displaying a table with the results of the query. The table has 7 columns: flight_id, flight_number, departure_airport, arrival_airport, departure_time, and arrival_time. The data shows one row with flight_id 601, flight_number FL-2024, departure from ALA, arrival at TSE, and times for May 10, 2024.

flight_id	flight_number	departure_airport	arrival_airport	departure_time	arrival_time
601	FL-2024	ALA	TSE	2024-05-10 10:00:00	2024-05-10 12:00:00

A green status bar at the bottom indicates: 'Successfully run. Total query runtime: 175 msec. 1 rows affected.'

6-task



The screenshot displays a PostgreSQL IDE interface. The top toolbar includes icons for file operations, query execution, and settings. The main editor area contains the following SQL script:

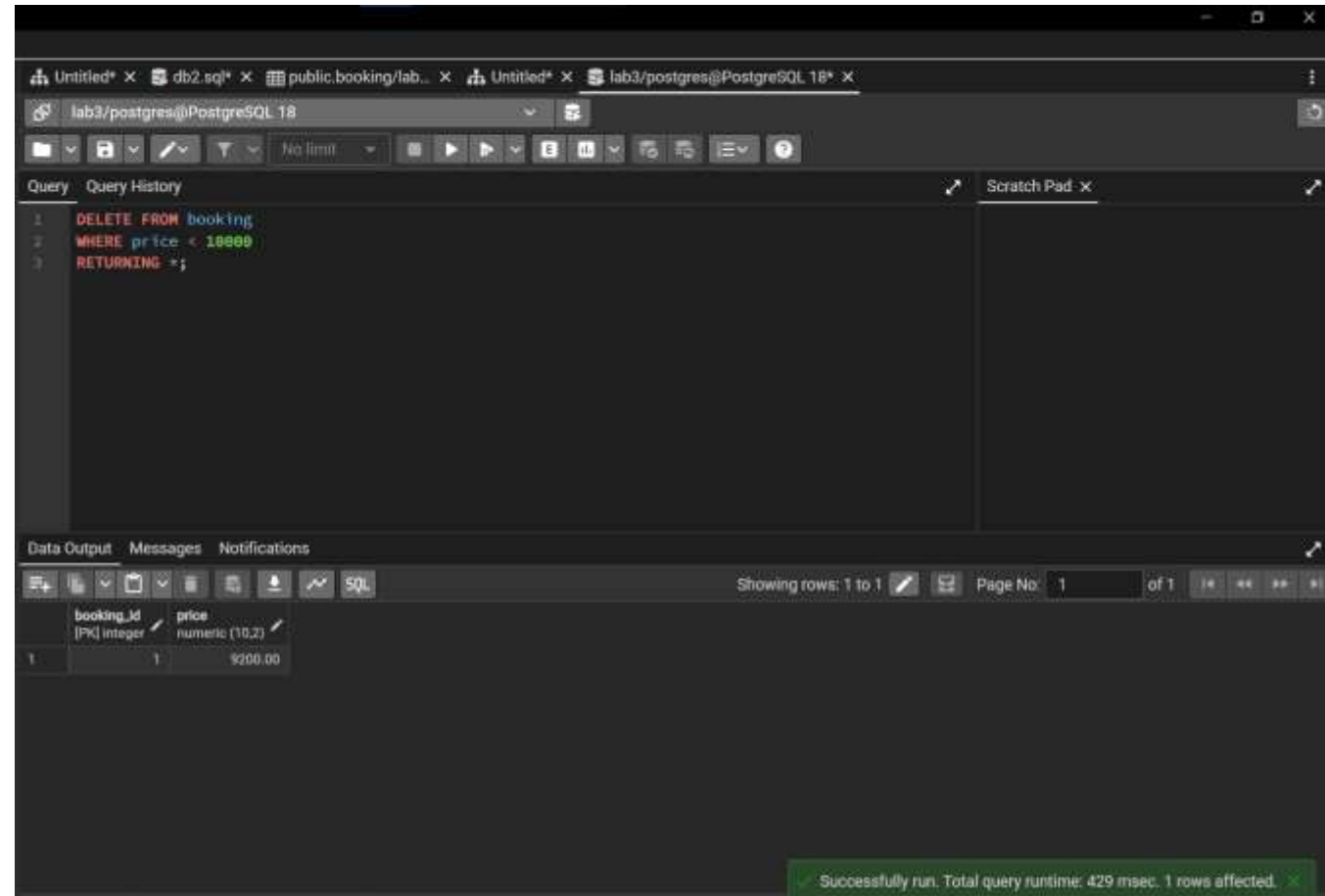
```
1 INSERT INTO booking (price) VALUES (8000), (12000), (15000);
2
3 UPDATE booking
4 SET price = price + 1.15
5 RETURNING *;
```

Below the editor, the 'Data Output' tab is active, showing the results of the query. The results are displayed in a table with two columns: 'booking_id' (integer) and 'price' (numeric). The table contains three rows of data.

booking_id	price
1	9200.00
2	13800.00
3	17250.00

A green status bar at the bottom indicates: 'Successfully run. Total query runtime: 240 msec. 3 rows affected.' The bottom status bar also shows 'Total rows: 3' and 'Query complete 00:00:00.240'.

7-task



The screenshot shows a PostgreSQL query editor interface. The top toolbar includes icons for file operations, query execution, and settings. The main query area contains the following SQL code:

```
1 DELETE FROM booking
2 WHERE price < 10000
3 RETURNING *;
```

Below the query editor, the 'Data Output' tab is active, displaying the results of the query in a table format. The table has two columns: 'booking_id' (integer, primary key) and 'price' (numeric, 10,2). The results show one row with 'booking_id' 1 and 'price' 9200.00.

booking_id	price
1	9200.00

At the bottom of the interface, a green status bar indicates: 'Successfully run. Total query runtime: 429 msec. 1 rows affected.'

8-task

The screenshot shows a PostgreSQL client window with the following components:

- Query Editor:** Contains the SQL query:

```
1 UPDATE airline
2 SET airline_code = 'UNK'
3 WHERE airline_code IS NULL
4 RETURNING *;
```
- Data Output:** Displays the results of the query in a table with 5 rows and 5 columns.
- Status Bar:** Shows "Total rows: 5" and "Query complete 00:00:00.232".
- Notification:** A green message box at the bottom right states: "Successfully run. Total query runtime: 232 msec. 5 rows affected."

	airline_id (PK) integer	name character varying (100)	country character varying (100)	airline_code character varying (10)	update_at timestamp without time zone
1	1	KazAir	Turkey	UNK	2026-09-27 11:26:06.523291
2	2	KazAir	Turkey	UNK	2026-09-27 11:36:19.56897
3	3	AirEasy	France	UNK	2026-09-27 11:37:06.645018
4	4	FlyHigh	Brazil	UNK	2026-09-27 11:37:06.645018
5	5	FlyFly	Poland	UNK	2026-09-27 11:37:06.645018

9-task

The screenshot shows a PostgreSQL client window with the following components:

- Tab Bar:** Contains several tabs, including 'lab3/postgres@PostgreSQL 18'.
- Toolbar:** Includes icons for file operations, query execution, and other database functions.
- Query Editor:** Displays the following SQL query:

```
1 INSERT INTO baggage_check (status, created_at) VALUES ('Not checked', '2023-01-15');
2
3 DELETE FROM baggage_check
4 WHERE created_at < '2023-06-01'
5 AND status = 'Not checked'
6 RETURNING *;
```
- Data Output Panel:** Shows the result of the query as a table with the following columns and data:

baggage_check_id	status	created_at
1	Not checked	2023-01-15 00:00:00
- Messages Panel:** Displays a green notification: 'Successfully run. Total query runtime: 282 msec. 1 rows affected.'

10-task

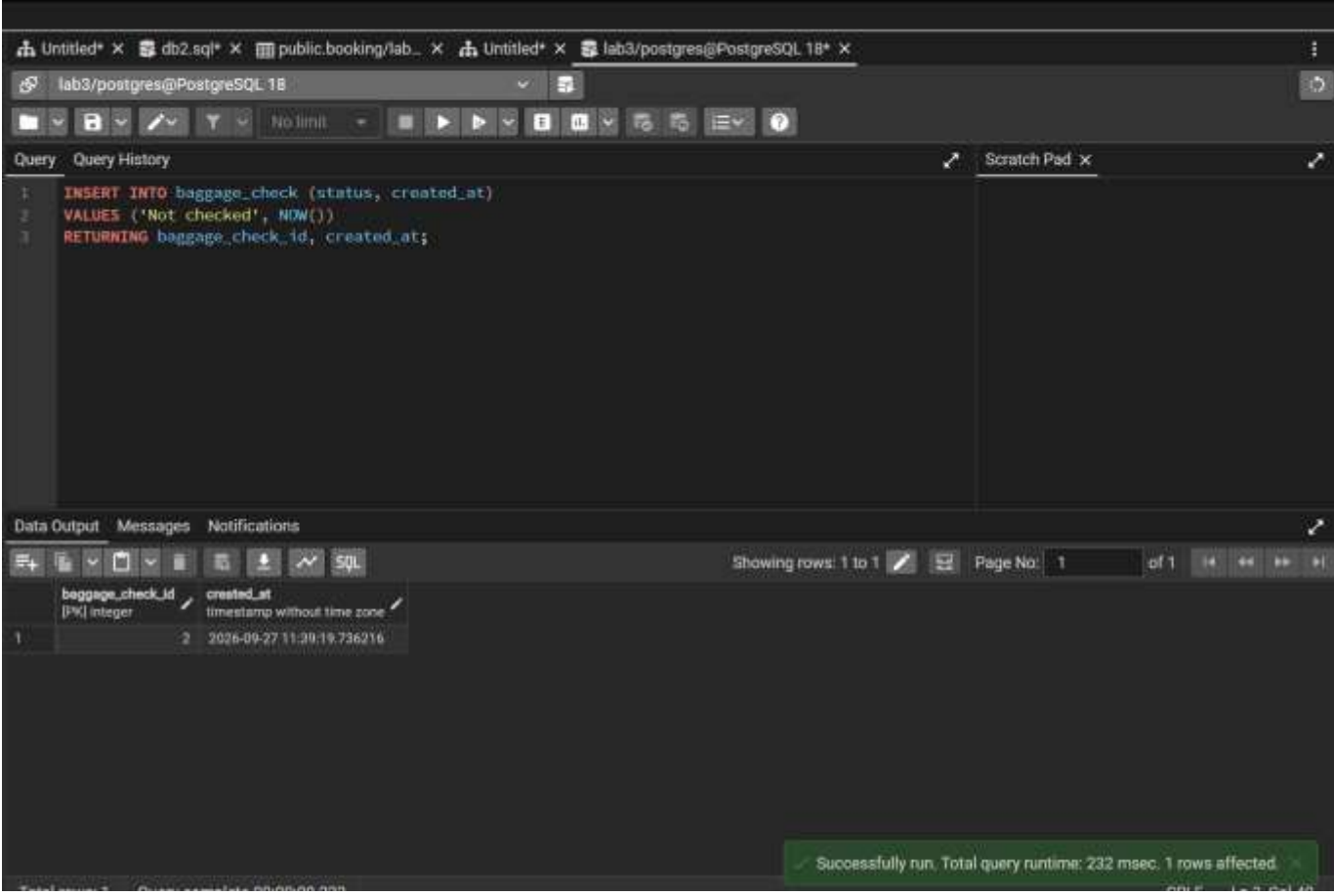
The screenshot shows a PostgreSQL IDE interface with the following components:

- Top Bar:** Displays the current database connection as 'lab3/postgres@PostgreSQL 18'.
- Query Editor:** Contains the following SQL script:

```
1 INSERT INTO airport (city, state) VALUES ('Mlawe', NULL), ('Kepuh', NULL);
2
3 DELETE FROM airport
4 WHERE state IS NULL
5 AND city IN ('Mlawe', 'Kepuh')
6 RETURNING *;
```
- Data Output Panel:** Shows the results of the query in a table format:

	airport_id [PK] integer	city character varying (100)	state character varying (100)
1	1	Mlawe	[null]
2	2	Kepuh	[null]
- Status Bar:** Indicates 'Total rows: 2' and 'Query complete 00:00:00.381'.
- Message Panel:** Displays a green notification: 'Successfully run. Total query runtime: 381 msec. 2 rows affected.'

11-task



The screenshot shows a PostgreSQL IDE interface with the following components:

- Top Bar:** Displays the current database connection as `lab3/postgres@PostgreSQL 18`.
- Query Editor:** Contains the following SQL query:

```
1 INSERT INTO baggage_check (status, created_at)
2 VALUES ('Not checked', NOW())
3 RETURNING baggage_check_id, created_at;
```
- Data Output Panel:** Shows the results of the query in a table format:

	baggage_check_id [PK] integer	created_at timestamp without time zone
1	2	2026-09-27 11:39:19.736216
- Status Bar:** A green message at the bottom indicates: "Successfully run. Total query runtime: 232 msec. 1 rows affected."

12-task

The screenshot shows a PostgreSQL IDE interface with the following components:

- Query Editor:** Contains the SQL query:

```
1 UPDATE airline;
2 SET country = UPPER(country);
3 RETURNING *;
```
- Data Output:** Displays the results of the query in a table with 5 rows. The columns are: `airline_id` (PK integer), `name` (character varying (100)), `country` (character varying (100)), `airline_code` (character varying (10)), and `update_at` (timestamp without time zone).
- Status Bar:** Shows "Total rows: 5" and "Query complete 00:00:00.211".
- Message:** A green notification bar at the bottom right states: "Successfully run. Total query runtime: 211 msec. 5 rows affected."

	airline_id (PK) integer	name character varying (100)	country character varying (100)	airline_code character varying (10)	update_at timestamp without time zone
1	1	KazAir	TURKEY	UNK	2026-09-27 11:26:06.523291
2	2	KazAir	TURKEY	UNK	2026-09-27 11:36:19.56597
3	3	AirEasy	FRANCE	UNK	2026-09-27 11:37:06.645018
4	4	FlyHigh	BRAZIL	UNK	2026-09-27 11:37:06.645018
5	5	FlyFly	POLAND	UNK	2026-09-27 11:37:06.645018

13-task

The screenshot shows a PostgreSQL query editor interface. The top toolbar includes icons for file operations, query execution, and settings. The main query editor contains the following SQL code:

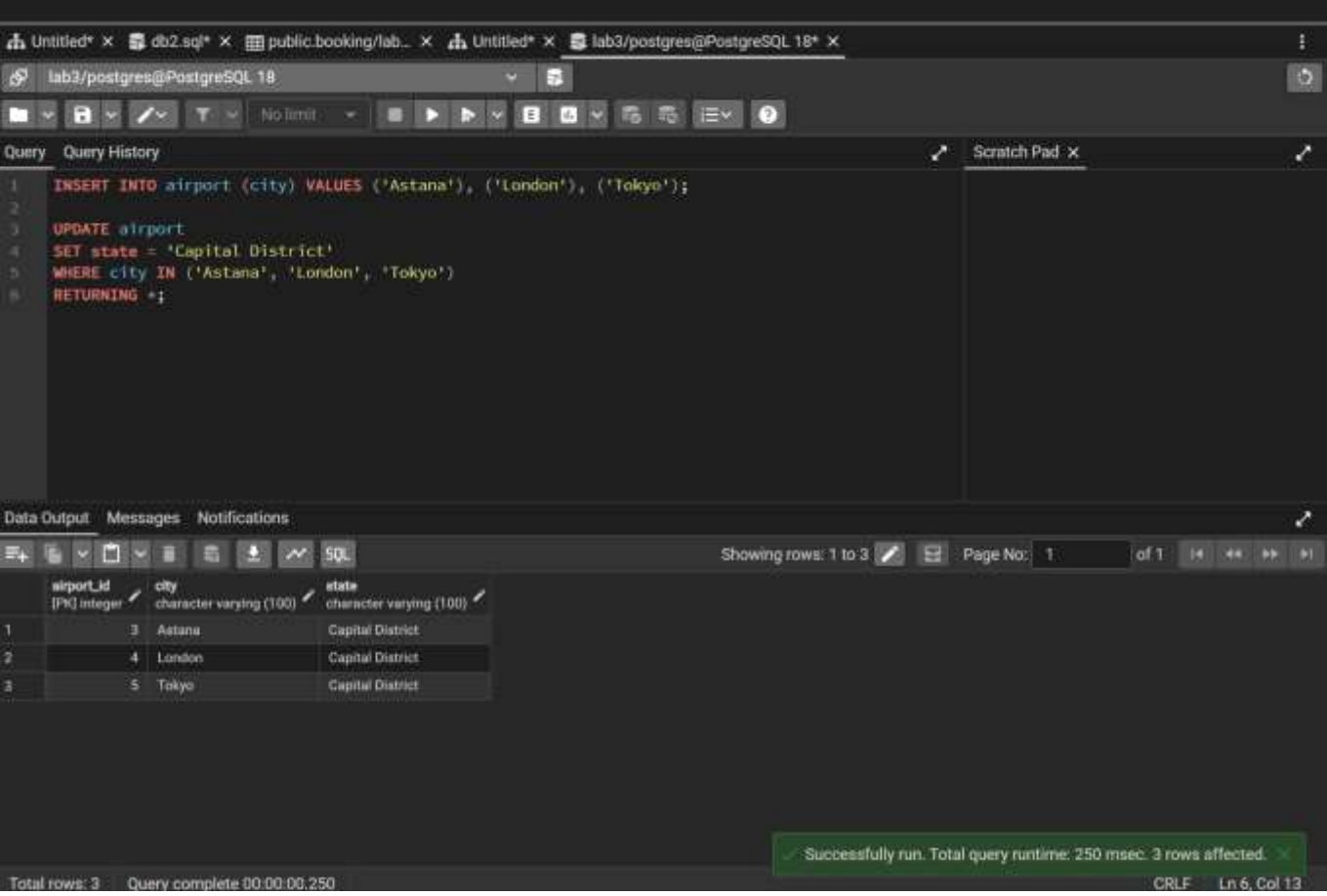
```
1 INSERT INTO airline (airline_id, name, country) VALUES (5, 'Old Name', 'Old Country')
2 ON CONFLICT (airline_id) DO NOTHING;
3
4 UPDATE airline
5 SET name = 'Global Airways',
6     country = 'United Kingdom',
7     update_at = CURRENT_DATE
8 WHERE airline_id = 5
9 RETURNING *;
```

Below the query editor, the 'Data Output' tab is active, displaying the results of the query in a table format. The table has five columns: `airline_id` (integer), `name` (character varying), `country` (character varying), `airline_code` (character varying), and `update_at` (timestamp without time zone). The results show one row with the following values:

airline_id	name	country	airline_code	update_at
5	Global Airways	United Kingdom	UNK	2026-09-27 00:00:00

At the bottom of the interface, a green status bar indicates: "Successfully run. Total query runtime: 355 msec. 1 rows affected."

14-task



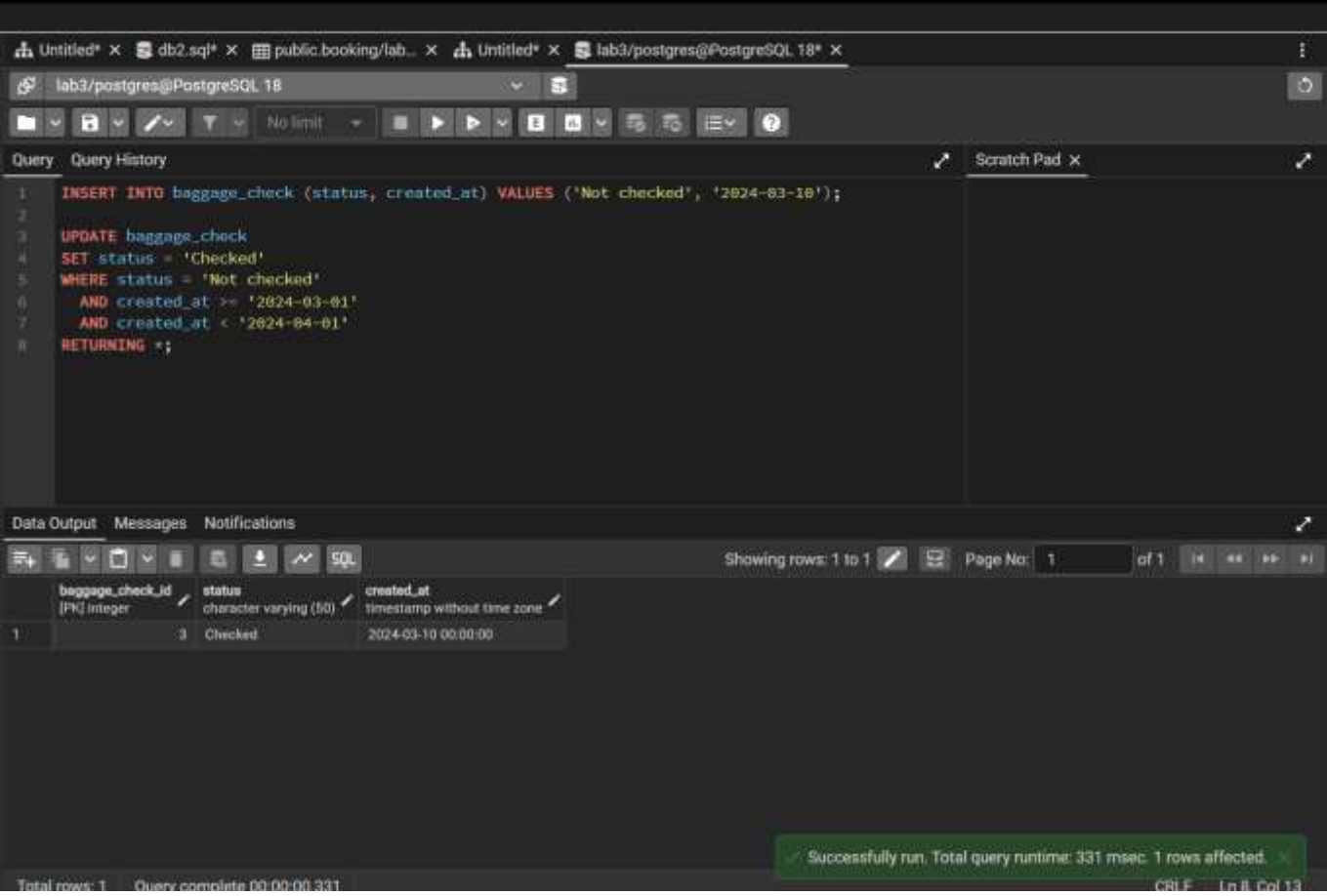
The screenshot shows a PostgreSQL IDE interface with the following components:

- Query Editor:** Contains the following SQL script:

```
1 INSERT INTO airport (city) VALUES ('Astana'), ('London'), ('Tokyo');
2
3 UPDATE airport
4 SET state = 'Capital District'
5 WHERE city IN ('Astana', 'London', 'Tokyo')
6 RETURNING *;
```
- Data Output:** Displays the results of the query in a table format:

	airport_id [PK] integer	city character varying (100)	state character varying (100)
1	3	Astana	Capital District
2	4	London	Capital District
3	5	Tokyo	Capital District
- Status Bar:** Shows "Total rows: 3" and "Query complete 00:00:00.250".
- Message:** A green notification box at the bottom right states: "Successfully run. Total query runtime: 250 msec. 3 rows affected."

15-task



The screenshot shows a PostgreSQL query editor interface. The top toolbar includes icons for file operations, query execution, and settings. The main editor area contains the following SQL query:

```
1 INSERT INTO baggage_check (status, created_at) VALUES ('Not checked', '2024-03-10');
2
3 UPDATE baggage_check
4 SET status = 'Checked'
5 WHERE status = 'Not checked'
6 AND created_at >= '2024-03-01'
7 AND created_at < '2024-04-01'
8 RETURNING *;
```

Below the query editor, the 'Data Output' tab is active, displaying the results of the query in a table format. The table has three columns: `baggage_check_id` (PK Integer), `status` (character varying (50)), and `created_at` (timestamp without time zone). The results show one row with the following values:

baggage_check_id	status	created_at
1	Checked	2024-03-10 00:00:00

At the bottom of the interface, a green status bar indicates: 'Successfully run. Total query runtime: 331 msec. 1 rows affected.' The footer shows 'Total rows: 1' and 'Query complete: 00:00:00.331'.